

OMM Lab #33 Pre-Lab Lecture:

Palpation of the Cranial Strain Patterns, Condylar Decompression and Lifts



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Do.
Make.
Heal.

Innovate.

Reinvent the Future.

Before getting started, be sure to watch/study the required material listed in the CPG:

- PPOM2 Lecture - Introduction to Osteopathy in the Cranial Field - Dr. Ettlinger
- PPOM2 Lecture - Sphenobasilar Synchondrosis (SBS) Strain Patterns - Dr. Yao
- PPOM2 OMM Lab #32 - Introduction to Cranial Diagnosis and Treatment - Landmarks, Vault Contacts, Strain Patterns, and OA decompression treatment
- OMM Lab #32 Prep lecture

Failure to understand the above material will put you at a huge disadvantage in this lab.

Session Objectives

- Be able to explain, using “air hand models,” what hand positions you would predominantly sense with each of the following cranial strain pattern diagnoses: flexion, extension, torsions, sidebending and rotation, superior shear, inferior shear, lateral strains, and compression, then try to sense what strain pattern is present in your partner(s)
- Begin to sense the strain patterns in their partner(s)
- Be able to discuss the indications and contraindications of treating patients with Osteopathic Cranial Manipulation
- Be able to perform condylar decompression (a.k.a. decompression of occipital condyles) and the Frontal, Parietal, Zygomatic lifts and the nasion spread
- Review material from PPOM2 Lab 32 and PPOM2 Lecture #13 (Dr. Yao’s lecture): Be able to utilize the vault contact and demonstrate each cranial strain pattern with “air models.” You should be able to discuss the number and direction of the axes of motion and motion of the sphenoid and occiput around the axes in relation to the axes and each other and discuss if the strain patterns are physiologic or non-physiologic.

Review of Physiologic Strain Patterns

Pattern	Number and Direction of Axis	Movement of Occiput & Sphenoid relative to each other	Physiologic or non-physiologic Pattern	Description of air hands
Flexion	2 transverse axes	Move around transverse axes in opposite directions	Physiologic	The operator's fingers move towards the feet, palms move laterally, fingers spread
Extension	2 transverse axes	Move around transverse axes in opposite directions	Physiologic	The operator's hands move cephalad, palms move medially, and fingers move slightly together
Torsion	1 A/P axis	Move around A/P axis in opposite directions	Physiologic	Hands rotate in opposite directions with the hand rotating cephalad on the side where Greater Wing of Sphenoid moves cephalad
Sidebending & Rotation	2 parallel vertical axes, 1 A/P axis	Sidebending: Move in opposite directions around parallel vertical axes Rotation: Move in same direction around A/P axis	Physiologic	Hands move caudally with fingers spreading out on side of convexity (direction of rotation) Hands move cephalad with fingers approximating on the side of concavity.

Review of Non-Physiologic Strain Patterns

Pattern	Number and Direction of Axis	Movement of Occiput & Sphenoid relative to each other	Physiologic or non-physiologic Pattern	Description of air hands
Lateral Strain	2 parallel vertical axes	Move around vertical axes in same direction	Non-physiologic	Index fingers move in direction of greater wing of sphenoid while 5 th fingers move to the opposite side. Named for the side of the pinkies
Superior Vertical Shear (Strain)	2 transverse axes	Move around transverse axes in same direction	Non-physiologic	Fingers point down towards feet. Hands rotate anteriorly.
Inferior Vertical Shear (Strain)	2 transverse axes	Move around transverse axes in same direction	Non-physiologic	Fingers point up toward head. Hands rotate posteriorly
Sphenobasilar Synchrondrosis Compression	No axes	N/A	Non-physiologic	Either feels rock-hard like a bowling ball or feels all the cranial patterns at once



Flexion



Extension



Right Torsion



Left Torsion



Right SB & Rot



Left SB & Rot



Right Lat. Strain



Left Lat. Strain



Superior Shear



Inferior Shear



Techniques:

Condylar decompression (a.k.a. decompression of occipital condyles)

- Direct technique
- Please see the lab handout and posted video. We will discuss in lab.

Frontal lift, Parietal lift, Zygomatic lift and Nasion spread

- Please see the lab handout and posted video. We will discuss in lab.

Indications for treating patients with Osteopathic Cranial Manipulative Medicine include:

- Headaches
- Mild to severe whiplash strain and sprain injuries
- Post-concussive syndromes/traumatic brain injury
- Vertigo and tinnitus
- Cranial neuropathy and/or nerve entrapment
- Trigeminal neuralgia
- Bell's palsy
- Otitis media with effusion and serous otitis media
- Temporomandibular joint dysfunction and malocclusion
- Sinusitis
- Colic
- Plagiocephaly
- Feeding difficulties

<https://meded-lwwhealthlibrary-com.arktos.nyit.edu/content.aspx?sectionid=252637062&bookid=3202>

Contraindications:

Absolute Contraindications (May Include but Not Limited to):

- Lack of patient consent
- Absence of somatic dysfunction
- Acute intracranial bleeding and hemorrhage (CVA)
- Acute skull fracture
- Certain seizure states (relative)

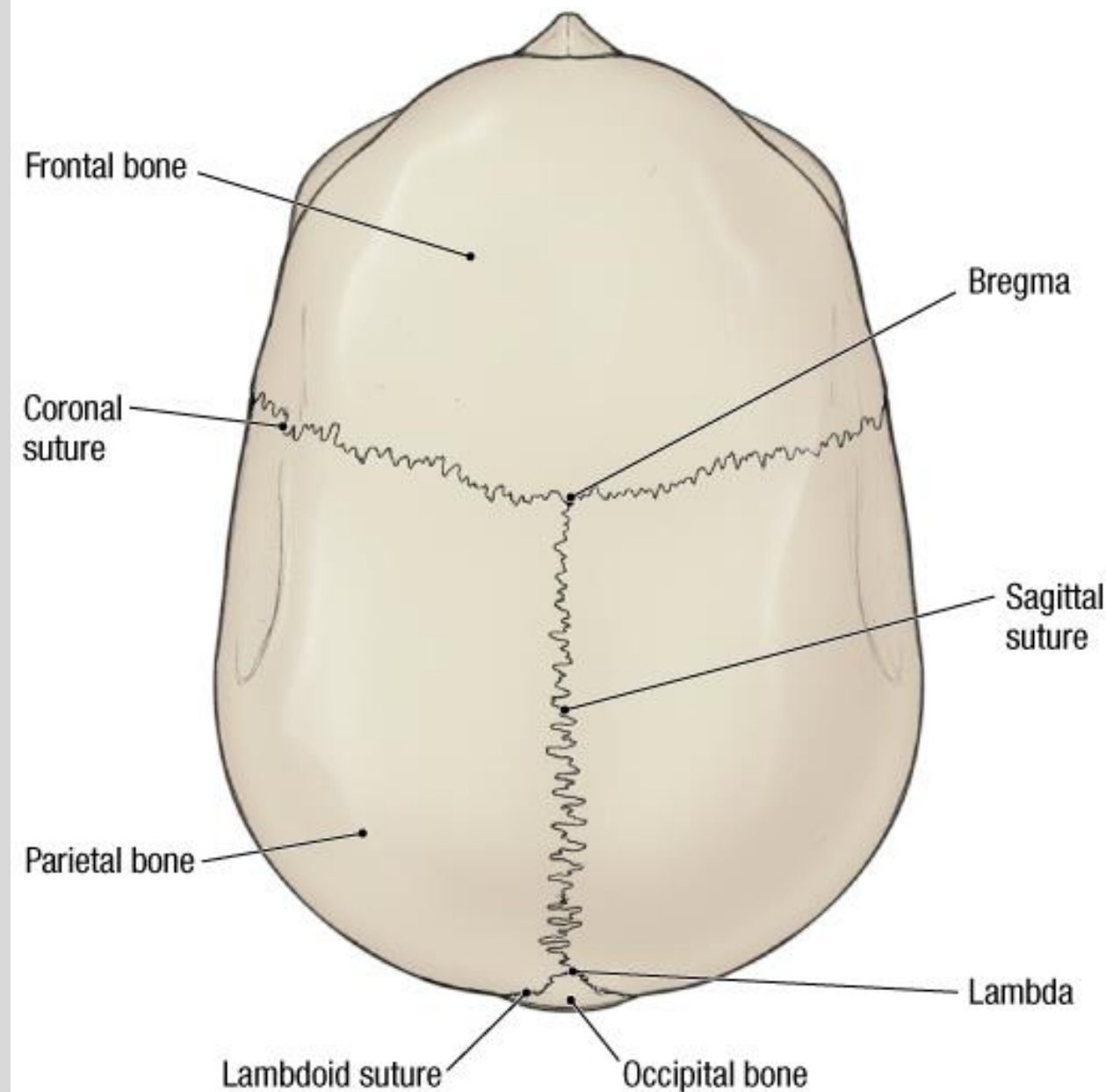
Relative Contraindications:

- Coagulopathies
- Space-occupying lesion in the cranium
- Increased intracranial pressure

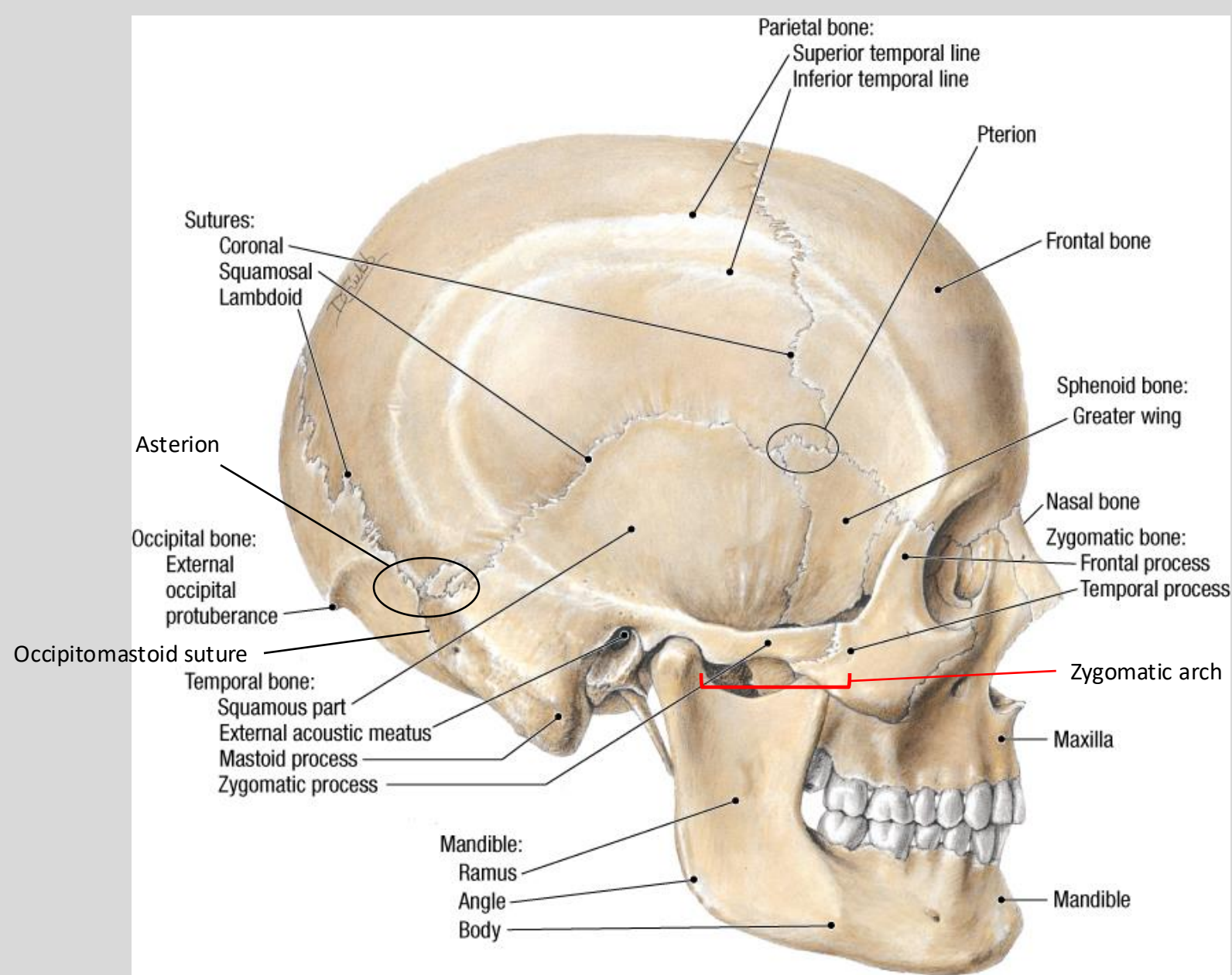
<https://meded-lwwhealthlibrary-com.arktos.nyit.edu/content.aspx?sectionid=252637062&bookid=3202>

Review Landmarks!

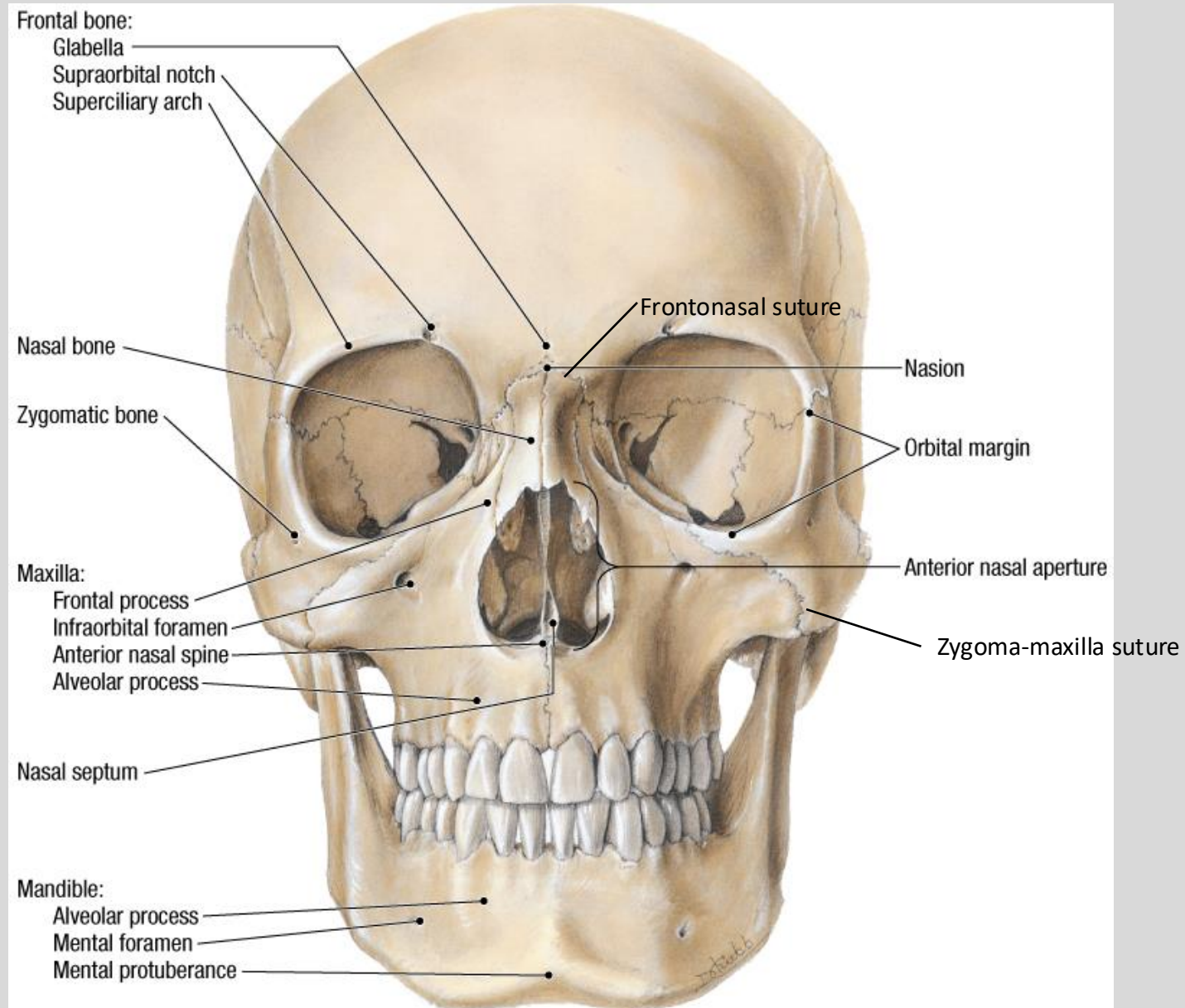
- **Nasion**
- **Nasal(s)**
- **Frontal(s)**: Zygomatic process of the frontal, Glabella, Bregma
- **Zygoma(e)**: temporal process, frontal process
- **Maxilla(e)**: zygoma-maxillary suture, frontal process
- **Parietal(s)**: Sutures (coronal, sagittal, squamous, lambdoid)
 - transverse sinus – **Asterion** – area of greatest amplitude in vault contact
- **Sphenoid**: greater wings, lesser wings
- **Temporal(s)**: zygomatic process, squama, mastoid process
- **Occiput**: inion (external occipital protuberance), lambda, nuchal lines, confluence of sinuses
- **Sutures**: coronal, sagittal, lambdoid, squamous, frontonasal, zygomatic arch, occipito- mastoid (O-M), frontal-lesser wing of sphenoid.
- **(Former) Fontanelles**: at Pterion, Asterion, Bregma and Lambda



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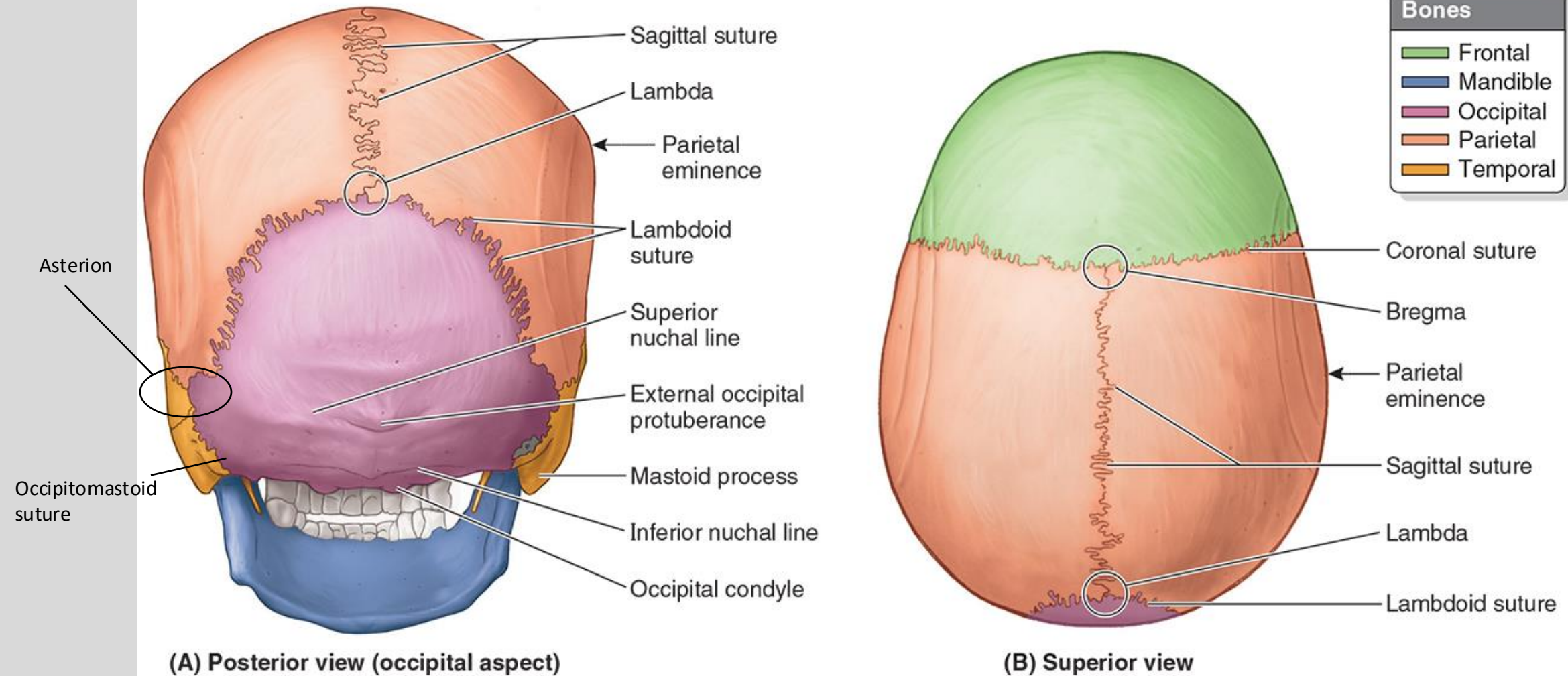
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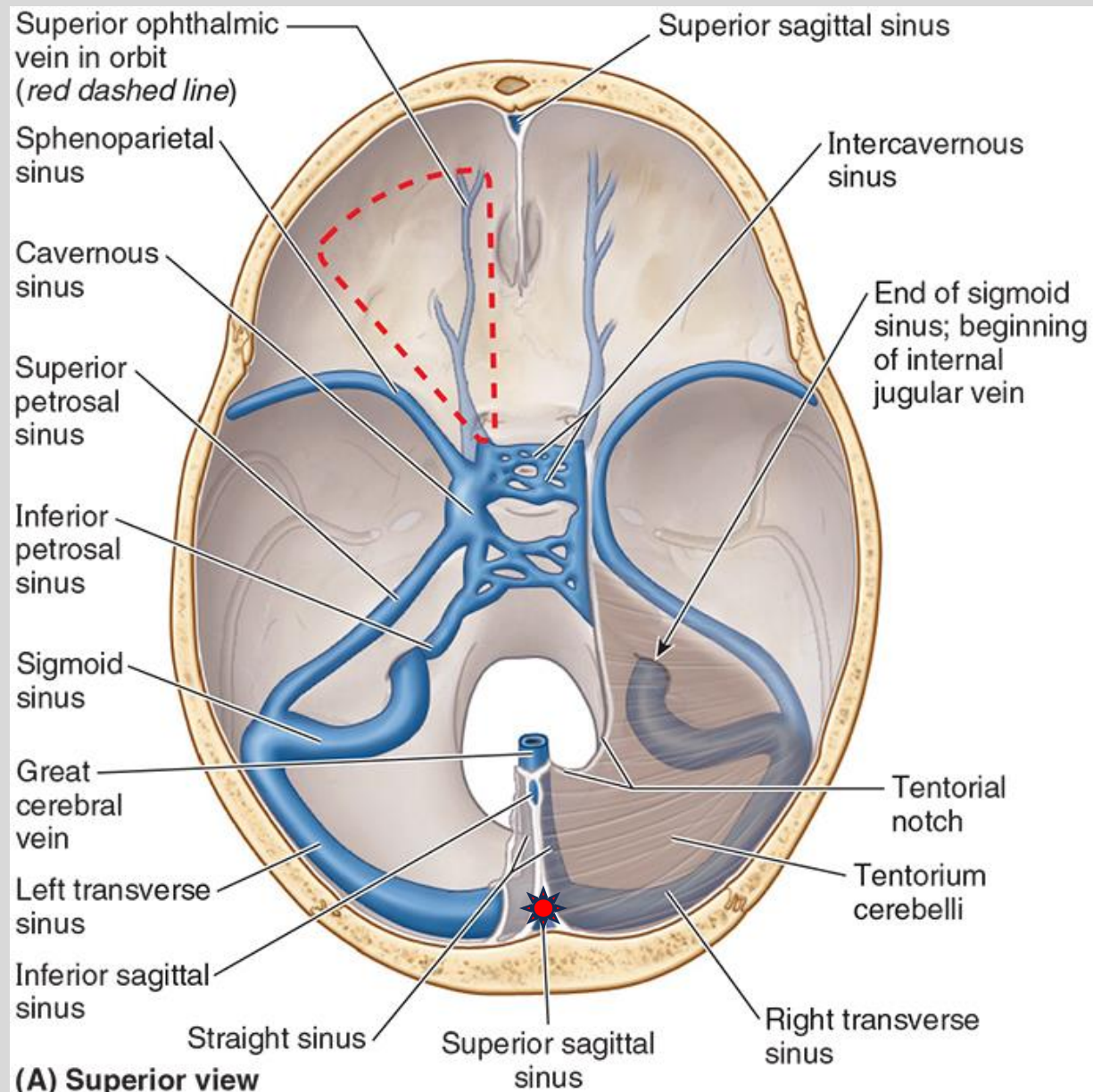
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Bones

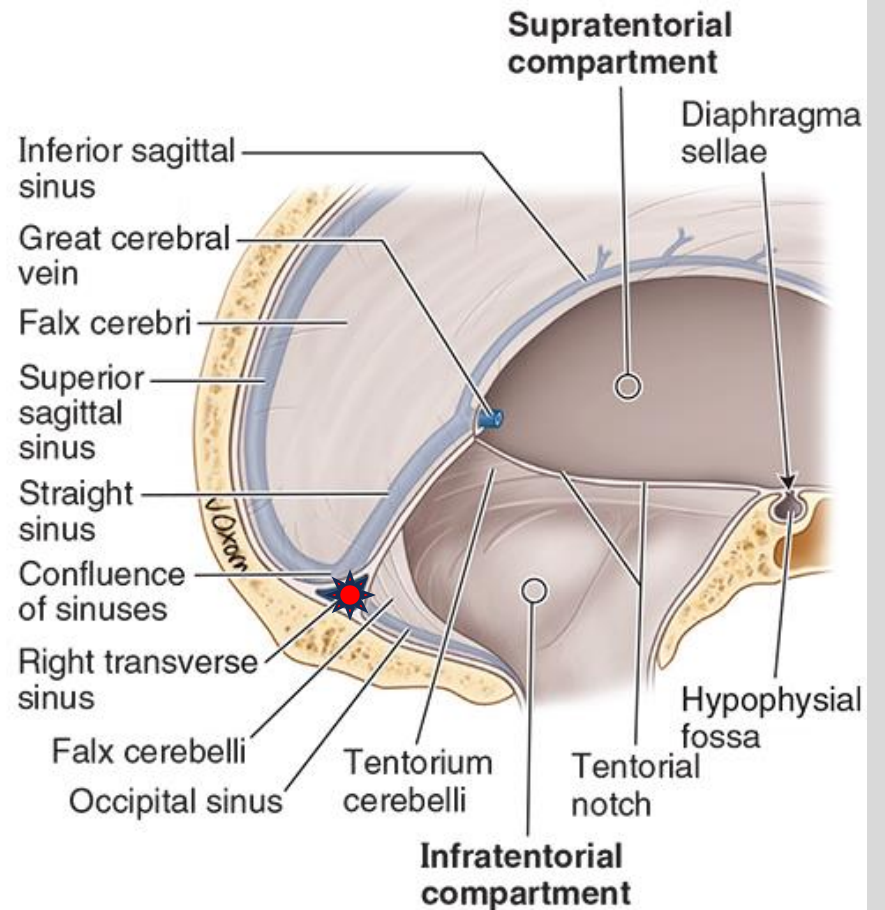
- Frontal
- Mandible
- Occipital
- Parietal
- Temporal



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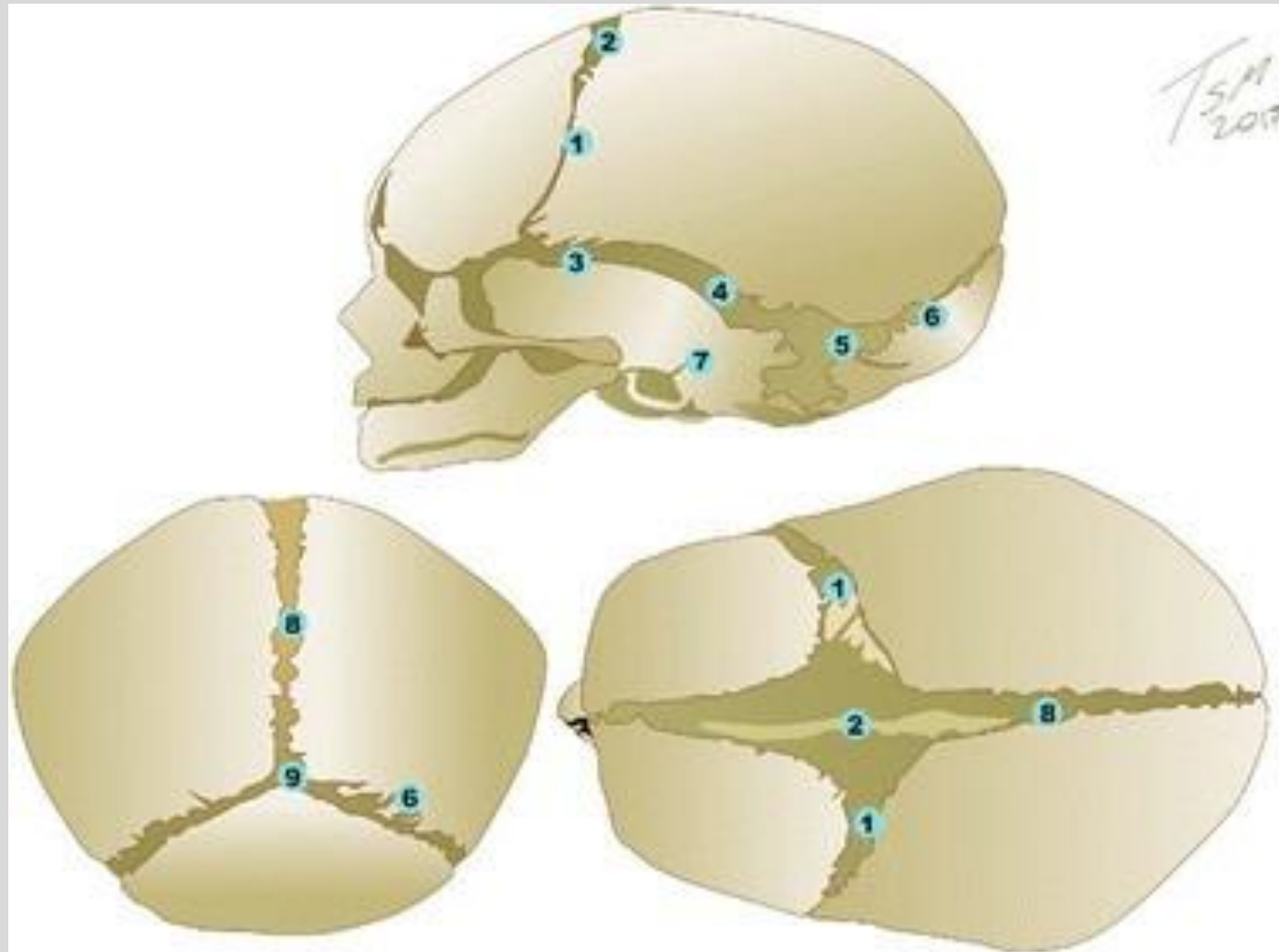
★ = Confluence of sinuses



(B) Medial view of left side

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Fetal Skull



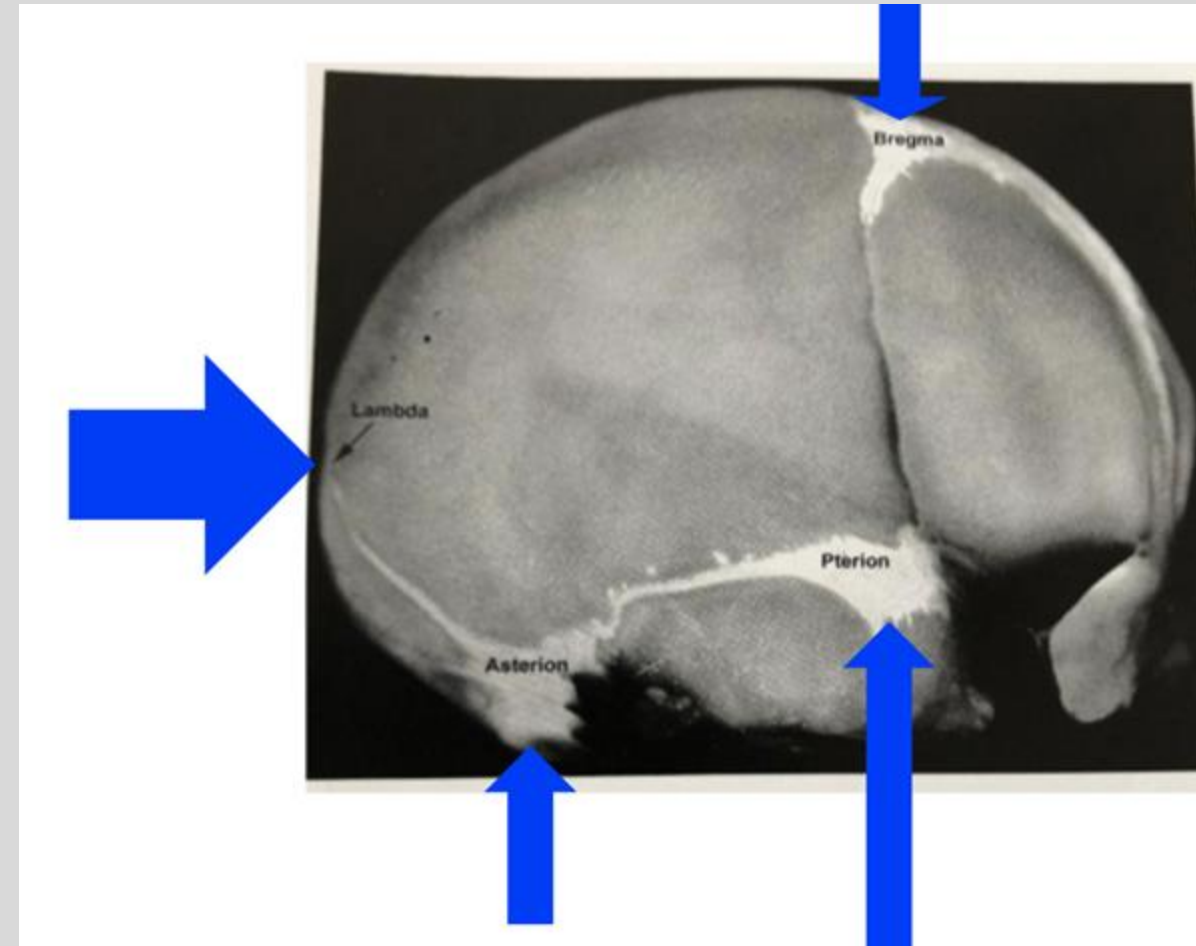
- 1, coronal suture; 2, anterior fontanelle; 3, anterolateral (sphenoid) fontanelle; 4, parietomastoid suture; 5, posterolateral fontanelle (occipitomastoid); 6, lambdoid suture; 7, external auditory meatus; 8, superior sagittal suture; 9, lambda.

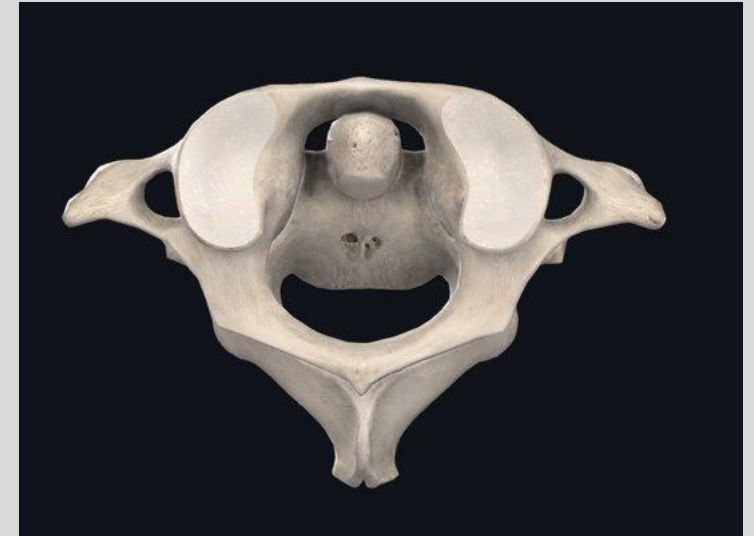
Infant Cranium

Six Fontanelles of the Infant

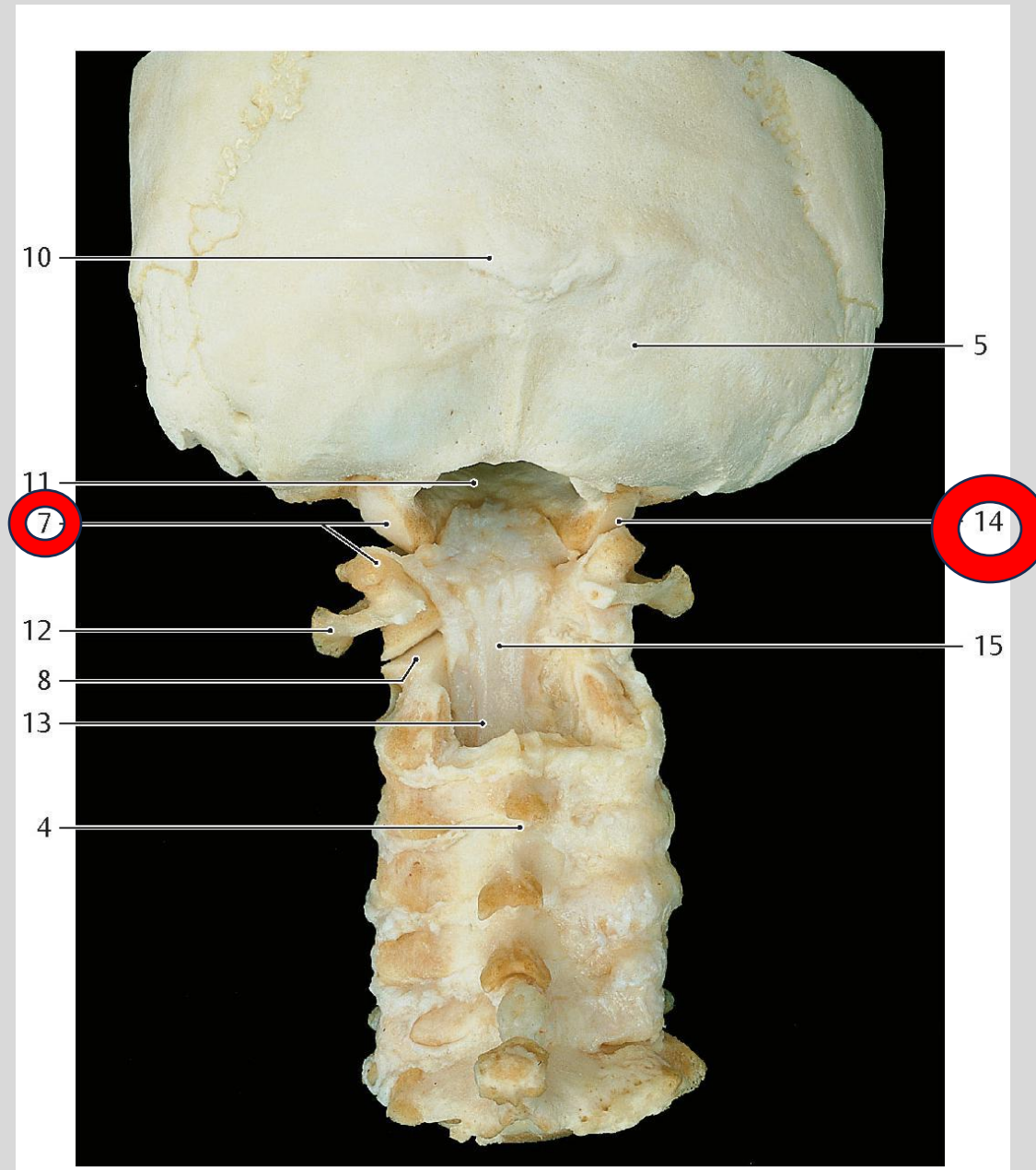
- Cranium in which the parietal bone is floating
- 29 cranial bones
 - 14 facial
 - 8 neurocranium
 - 6 ossicles
 - 1 hyoid
- 7 bones of the orbit:
- Frontal, zygomatic, maxillary, sphenoid, ethmoid, palatine, lacrimal
- Cranial vault is made up of membrane
- Cranial base is made up of cartilage
- 143 craniofacial articulations
- Teeth are not bone, but enamel is ectoderm

II, A. F. D., & Agur, A. M. R. (n.d.). Grant's Atlas of Anatomy, 14e. In *Grant's Atlas of Anatomy, 14e*, Chapter 7: Head. essay, Lippincott, Williams & Wilkins. LINK: <https://meded-lwwhealthlibrary-com.arktos.nyit.edu/content.aspx?sectionid=115777743&bookid=1737&resultClick=1#115777750>.





3D4 Medical



Condylar Decompression

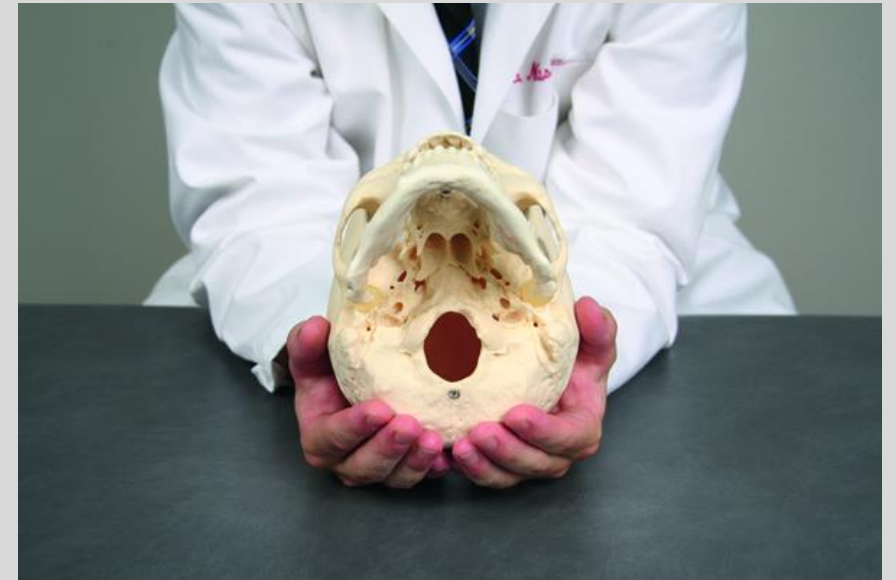
Landmarks:

- Suboccipital sulcus

Technique: See OMM Lab Video

Some Clinical Indications:

- Headache
- Suckling issues
- Colic
- Neck pain
- Low back pain



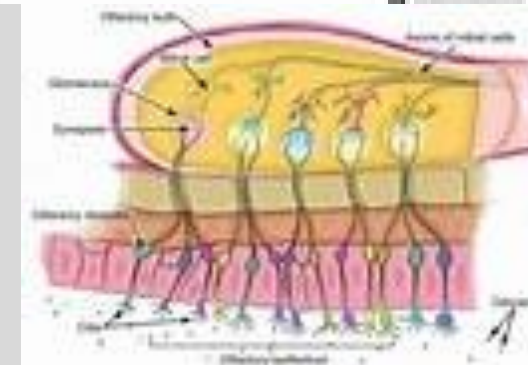
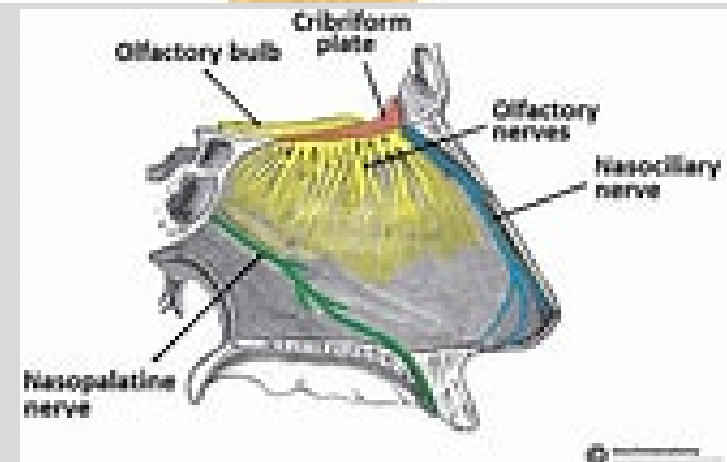
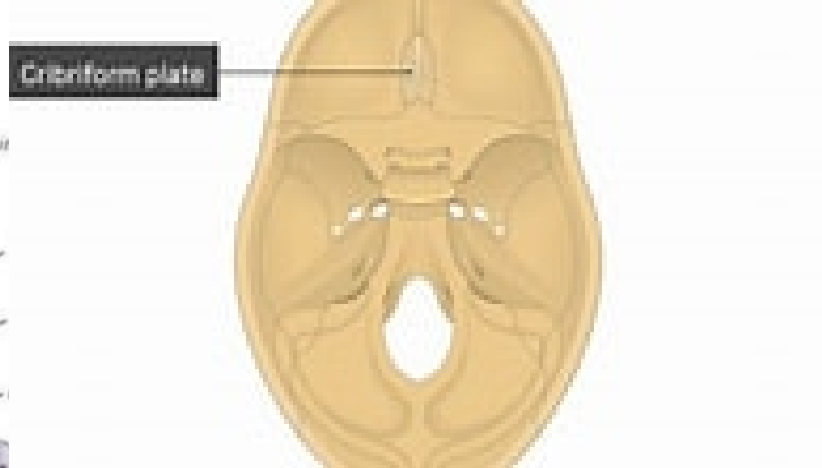
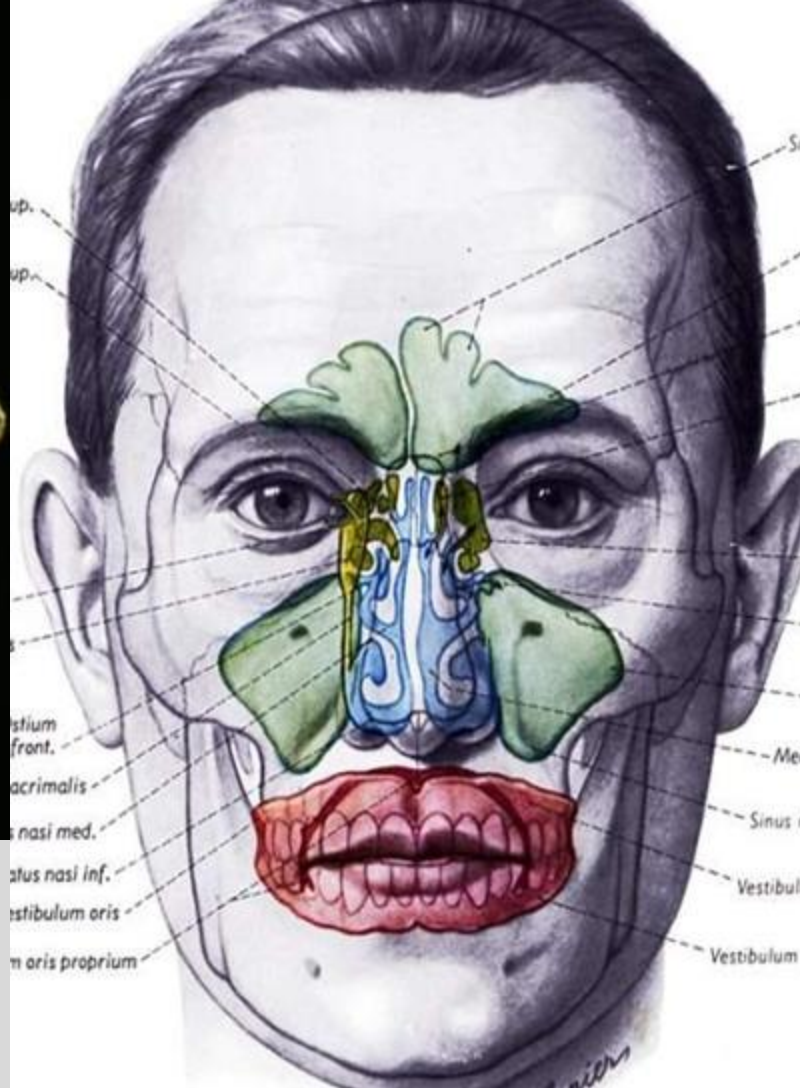
Atlas of Osteopathic Techniques, 4e, 2023 Copyright © Wolters Kluwer

Nicholas, Alexander S, and Evan A. Nicholas. *Atlas of Osteopathic Techniques*. Chapter 18: Osteopathic Cranial Manipulative Medicine, Philadelphia: Wolters Kluwer Health/Lippincott Williams & Wilkins, 2008. Link:

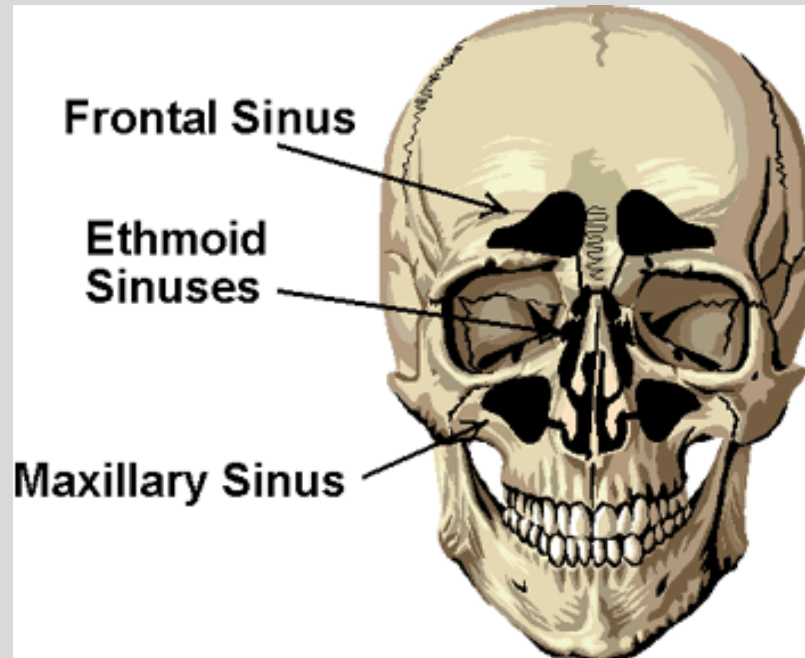
☀ = Ethmoid notch



Inferior Surface of Frontal Bone



The ethmoid notch freedom may impact effective function of the ethmoid's cribriform plate and healthy drainage of the frontal and ethmoid sinuses



Frontal Lift- Version 1

Landmarks:

- Frontozygomatic suture (superior to it)
- Coronal suture (anterior to it)

Technique: See OMM Lab Video

Some Clinical Indications:

- Frontal sinusitis/congestion
- Ethmoid bone restriction – dura attaches to crista galli
- Headaches/ Trauma/ Concussions/ Eye Complaints
- Frontal lobe mood/ executive function etc..
- Smell



Nicholas, Alexander S, and Evan A. Nicholas. *Atlas of Osteopathic Techniques*. Chapter 18: Osteopathic Cranial Manipulative Medicine, Section 13. Philadelphia: Wolters Kluwer Health/Lippincott Williams & Wilkins, 2008.

Nasion Spread

Landmarks:

- Nasal and frontal bones (If you go too far lateral you will be on the frontal processes of maxilla)

Technique: See OMM Lab Video

Clinical Indications:

- Sinusitis frontal (& ethmoid bone lies deep to nasion)
- Postnasal drip
- History of trauma, headaches, concussion
- Vision/ Eye complaints
- Sacral SD – (via dural connections to ethmoid)
- Anosmia/ Hyposmia/ Parosmia

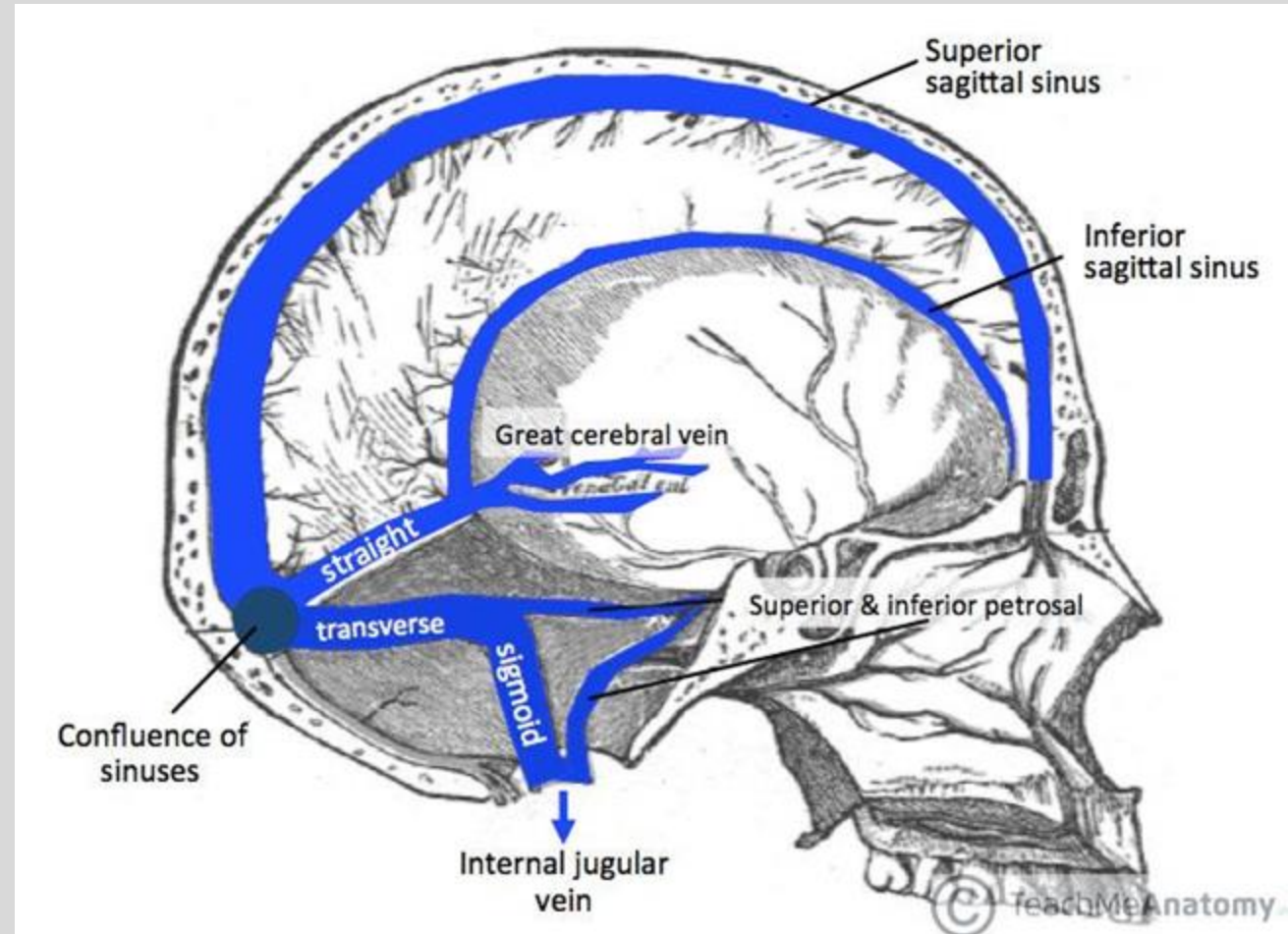
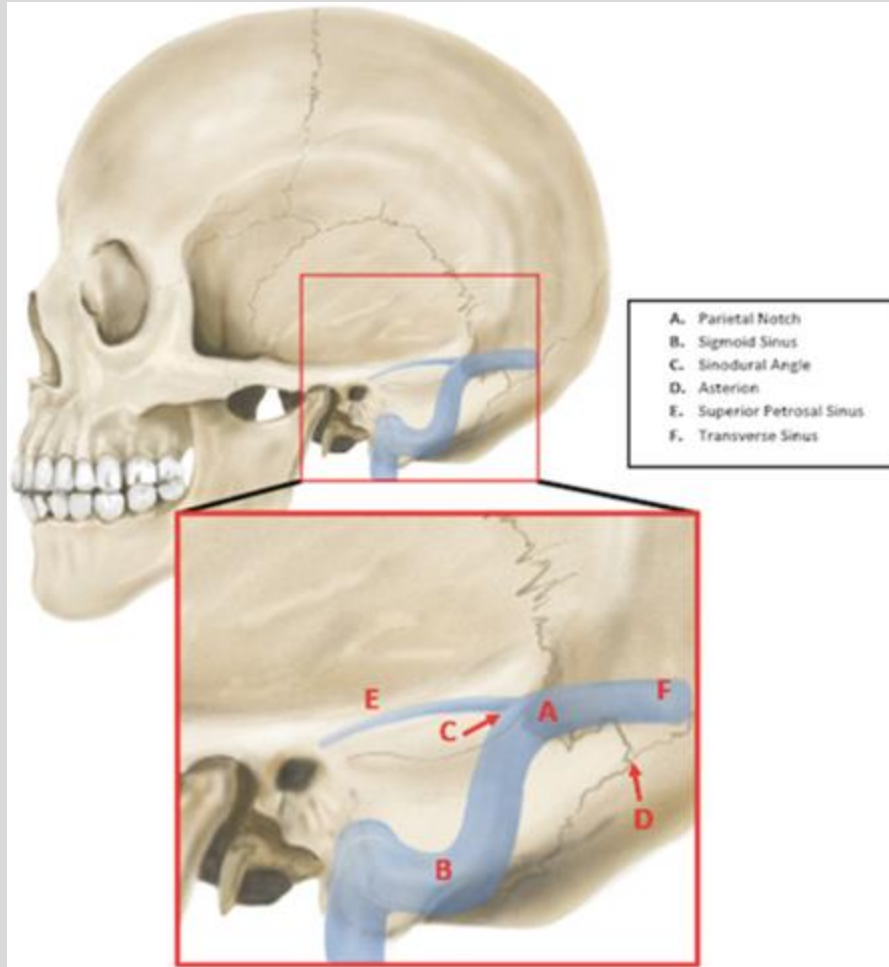


Nicholas, Alexander S, and Evan A. Nicholas. *Atlas of Osteopathic Techniques*. Chapter 18: Osteopathic Cranial Manipulative Medicine, Section 13. Philadelphia: Wolters Kluwer Health/Lippincott Williams & Wilkins, 2008.

Structure & Function: Venous Sinuses & Parietal Notch

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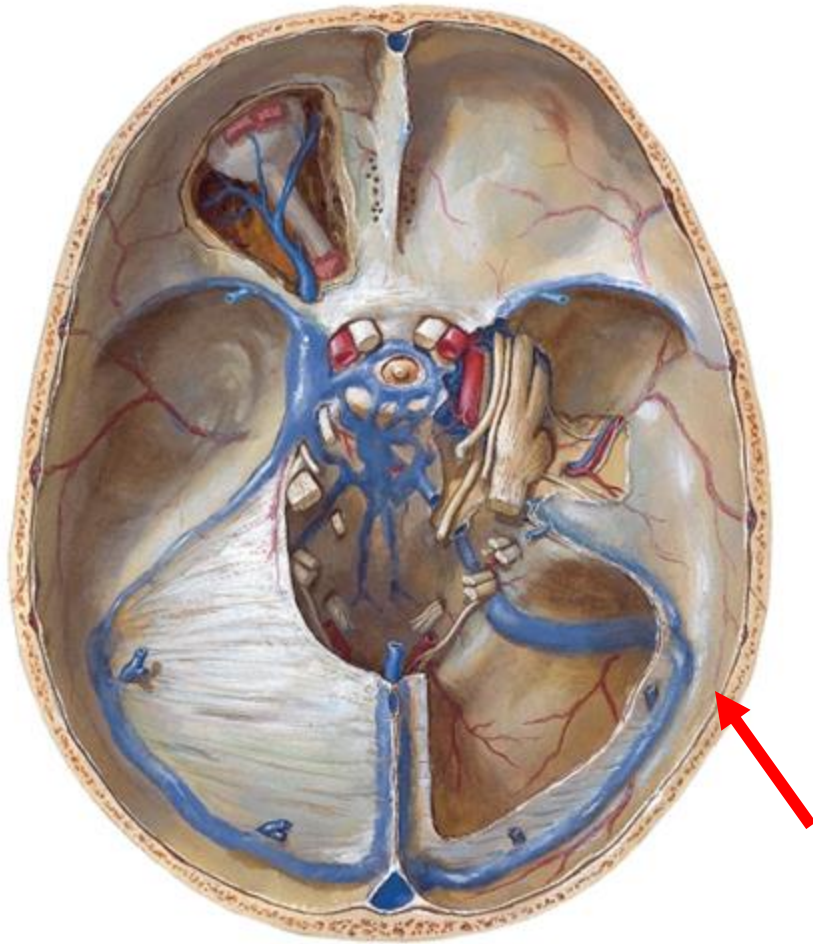
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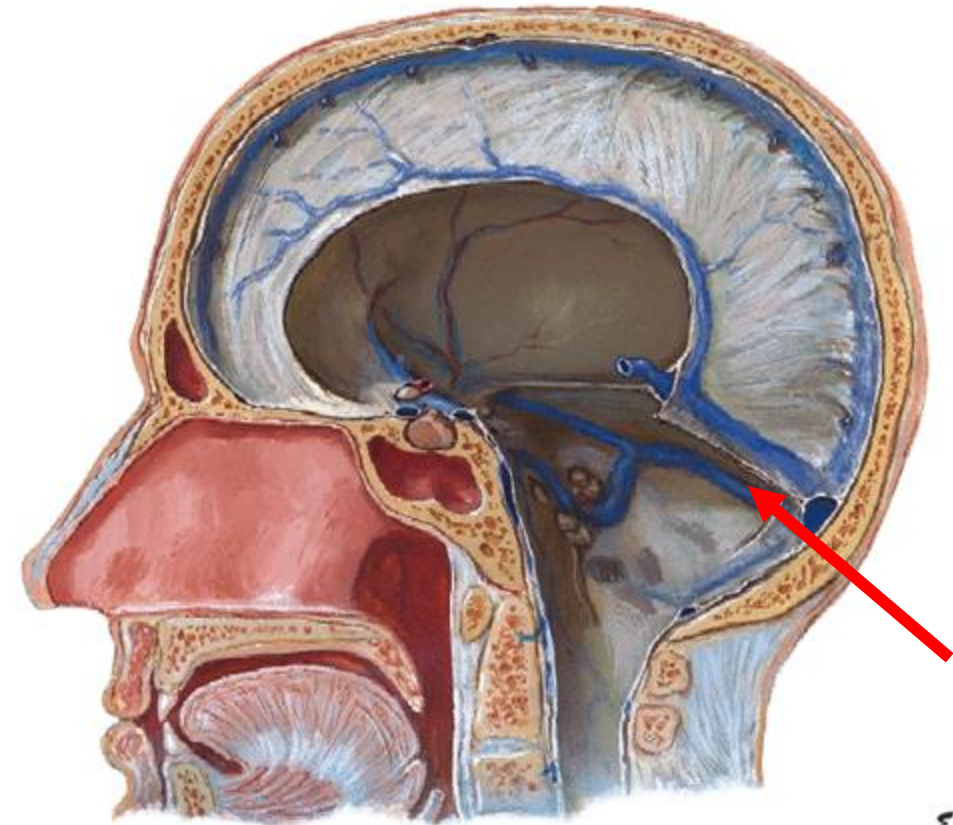
Castle, M., Myers, O. and Pickett, B., 2022. The parietal notch (Brammer's pointer): Accuracy of a surface landmark for temporal bone surgery. *Laryngoscope Investigative Otolaryngology*.

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Dural Venous Sinuses
Cranial Floor - Superior View



Dural Venous Sinuses
Sagittal section



- Parietal Lift: The transverse sinus and tentorium cerebelli cross near asterion.

Parietal Lift

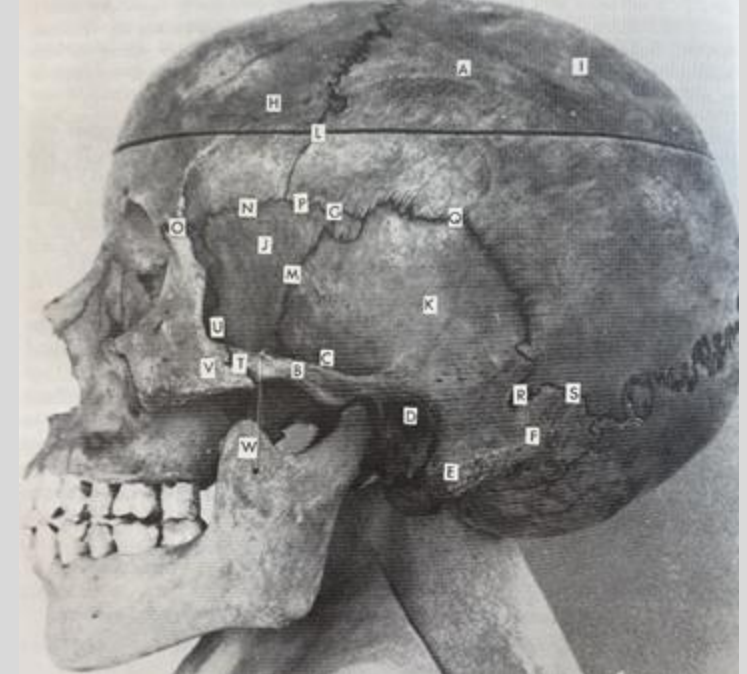
Landmarks:

- Sutures: Coronal, Squamous, Lambdoid & Asterion
- Bones: Frontal, parietal, temporal, occipital

Technique: See OMM Lab Video

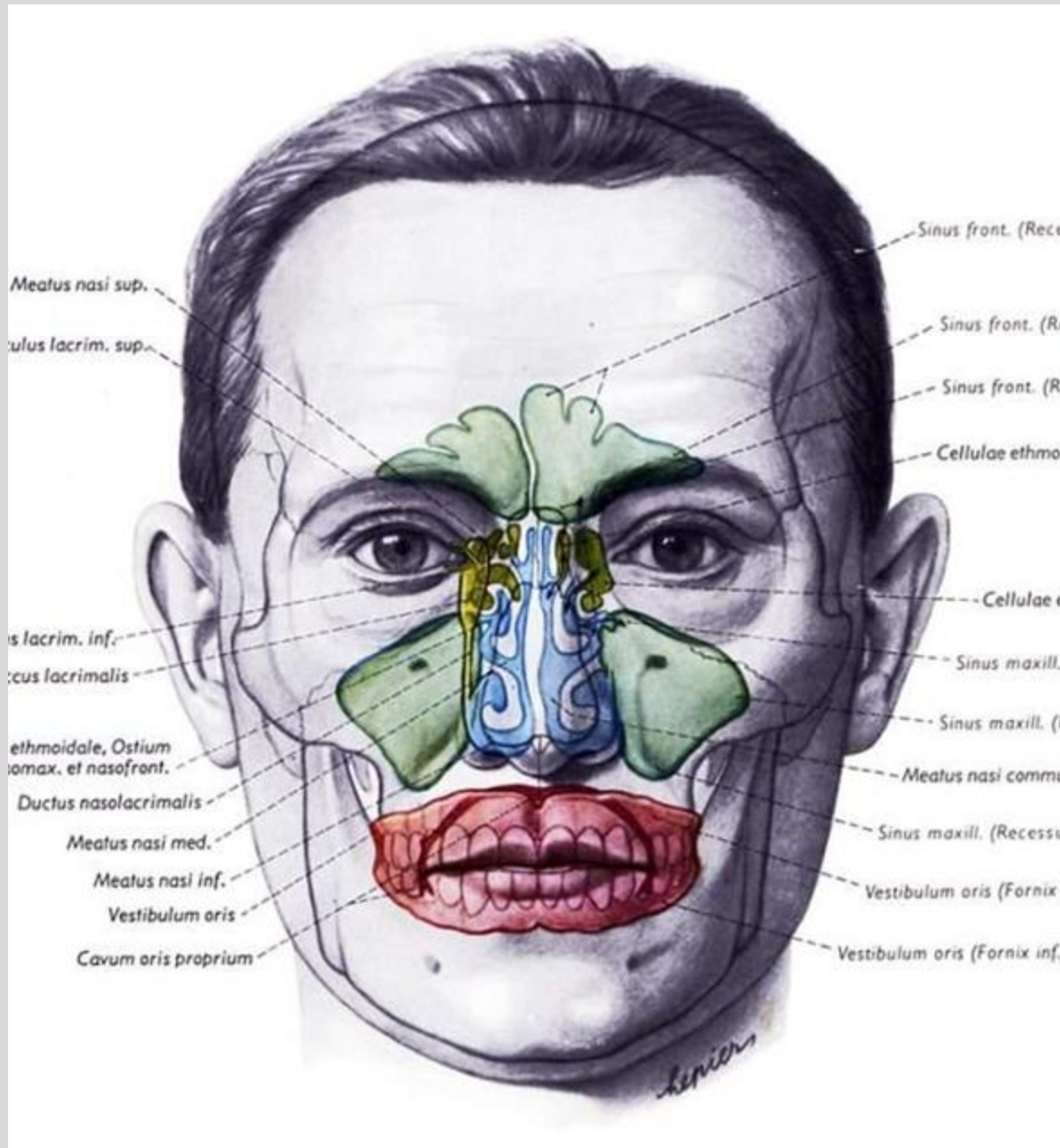
Some Clinical Indications:

- parietal restrictions involved in HA
- facial and venous sinus issues
- developmental abnormalities (all ages)



Sinuses

- The physiology of the nasopharyngeal region is also dependent on the motion within the bones of the face
- Exchange of air in the sinuses, as well as the drainage of secretions depends on normal movement of the facial mechanism



Sinuses

- The maxillary sinus is the most common site of sinusitis, because the opening of the sinus is high up on the medial side wall of the sinus



Image : pocketdentistry.com

Sinuses

Medicine



Frontal
Maxillae

Photo: courtesy Andrew Haltof, DO

- The maxilla “hangs” from the frontal
- Its axis of motion runs vertically (oblique) through the frontal process and through the body
- During inhalation, the midline of the face moves posteriorly and the lateral parts of the maxillae move laterally

Skull: Anterior View



Anterior Surface of Right Zygoma
Photo: courtesy Andrew Haltorf, DO

- The Zygoma moves about an oblique axis, influenced by the great wing of the sphenoid. Its medial border moves medially during inhalation – assisting with ‘pumping’ the antrum of the maxilla



Maxillae
Zygomae
Sphenoid

Photo: Courtesy Dr Andrew Haltof, DO

- The pump of the maxillary sinus is produced by the conflicting movement between the maxilla and zygoma
- Zygomatic (Maxillary) Lift

Zygomatic Lift

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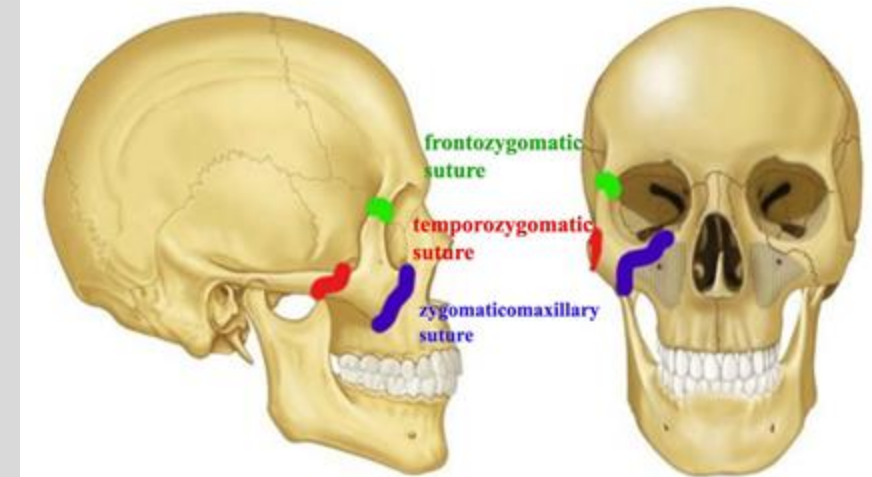
Landmarks:

- Zygomaticotemporal suture (anterior to it)
- Zyg-Max suture

Technique: See OMM Lab Video

Clinical Indications:

- Sinusitis (Maxillary Sinus)
- Vision
- Dental
- Trauma, HA's, Temporal Disorders, etc.



SCTF (Sutherland Cranial Teaching Foundation) and OCA (Osteopathic Cranial Academy) Course materials. Nicholas, Alexander S, and Evan A. Nicholas. *Atlas of Osteopathic Techniques*. Chapter 18: Osteopathic Cranial Manipulative Medicine, Section 13. Philadelphia: Wolters Kluwer Health/Lippincott Williams & Wilkins, 2008

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Case:

A 28-year-old female presents to your office complaining of headaches associated with stress at work over the past 6 years. She is a nurse and the practice she works at has been short staffed for the past 5 months. The headaches usually occur on both sides of her “temples” when at work and are sharp and around her head and last about 1-2 hours. The severity ranges from 3/10 to 8/10. She denies any photophobia or phonophobia, slurred speech or facial tingling or numbness. No visual changes. Rest and ibuprofen help the pain. She has seen a neurologist who told her she has tension headaches. She was told to take an antidepressant but does not want to take medications. Biofeedback and cognitive behavioral therapy are not covered by her insurance. She wants to know if OMT will help.

Case:

Past medical and surgical history: none

Medications: none

Allergies: Seasonal

Social History: denies recreational drug use, tobacco use; Drinks a glass of wine a few nights a week; does not exercise

Family History: Mother has type 2 DM

Review of systems: negative other than the headaches. Denies any dizziness, numbness, tingling or weakness.

Case:

Physical Exam:

Head: normocephalic/atraumatic

Eyes: EOMI, pupils equally round and reactive to light and accommodation

Ears: Tympanic membranes intact

Nose: mildly erythematous turbinates

Throat: slight post nasal drip.

Neck: supple, no masses or lymphadenopathy.

Neuro: CN II-XII grossly intact; stable gait; UE and LE motor 5/5 B/L, no gross sensory deficit; DTRs C5,C6, C7, L4, S1 all 2/4 B/L

Case:

Osteopathic Structural Exam:

Head forward, shoulders protracted

T1-6 Paravertebral muscle spasm (PVMS) (moderate)

B/L scapulae restriction (moderate), Bilateral pectoral spasm (moderate)

Right clavicle abducted (moderate), B/L rib 1 restriction R>L (moderate)

B/L trapezius spam R>L (moderate/severe)

C2 ERS right (severe), C2-5 PVMS (moderate)

OA compression (severe), frontonasal compression

Case:

Assessment:

1. Tension Headache
2. Somatic dysfunction Head
3. Somatic dysfunction Cervical region
4. Somatic dysfunction Ribs
5. Somatic dysfunction Thoracic
6. Somatic dysfunction Upper extremity

Case:

Plan:

1. Tension Headache – OMT, start exercising (walk 30 min a day to start), meditation/prayer
2. Somatic dysfunction Head – condylar decompression, frontal lift, zygomatic lift, nasion spread
3. Somatic dysfunction Cervical region – Soft tissue, MET, Balanced ligamentous tension (BLT)
4. Somatic dysfunction Ribs – thoracic outlet release, BLT
5. Somatic dysfunction Thoracic – soft tissue
6. Somatic dysfunction Upper extremity – BLT for clavicle and scapulae, pectoral traction
7. Stay well hydrated. Explained she may feel sore in the next day or two
8. Taught cervical spine stretches, pectoral stretches to be done 3 sets of 20 seconds with a few seconds rest in between, twice a day – stretch to stretch, not pain; rhomboid strengthening exercises to be done daily, 5-6 times a week.

Summary Slide

1. Reviewed cranial strain patterns and palpated them on your partner
2. Discussed the indications and contraindications of treating patients with Osteopathic Cranial Manipulation
3. Discussed specific indications for treating patients with condylar decompression (a.k.a. decompression of occipital condyles) and the Frontal, Parietal, Zygomatic lifts and the nasion spread
4. Practiced performing condylar decompression and the various lifts listed above

General References

- [Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e - Ch 38](#)
- [Atlas of Osteopathic Techniques, 4e - Ch 18](#)

Lecture and Lab Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

